Entropic Balog-Szemeredi-Gowers inequality

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Apache-2.0

Editorial note. Source-organized author material only. All mathematical source spans are retained unchanged. Blueprint lean/proves/uses metadata is hidden as formatting only; numbering is local to this material wrapper.

Editorial note. Ordinary cited prerequisites and concise author arguments are accepted as such. No independent mathematical certification or public ID allocation is claimed.

Allowed standard prerequisites (editorial).

- Standard finite Shannon entropy identities, conditional entropy and finite abelian group algebra are cited in their original Blueprint formulations.
- Named prior PFR/entropy or BSG results are allowed prerequisites; their original statements are retained where used, without demanding recursively expanded proofs.

1 Shannon entropy inequalities

Random variables in this paper are measurable maps $X : \Omega \to S$ from a probability space Ω to a measurable space S, and called S-valued random variables. In many cases we will assume that singletons in S are measurable. Often we will restrict further to the case when S is finite with the discrete σ-algebra, which of course implies that S has measurable singletons.

Definition 1 (Entropy)**.** *If X is an S-valued random variable, the entropy $\mathbb{H}[X]$ of X is defined*

$$\mathbb{H}[X] := \sum_{s \in S} \mathbb{P}[X = x] \log \frac{1}{\mathbb{P}[X = x]}$$

with the convention that $0 \log \frac{1}{0} = 0$.

Lemma 2 (Entropy and relabeling)**.** *(i) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ for some injection $f : S \to T$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.*

(ii) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ and $X = g(Y)$ for some functions $f : S \to T$, $g : T \to S$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.

Proof. Expand out both entropies and rearrange. □

We use X, Y to denote the pair $\omega \mapsto (X(\omega), Y(\omega))$.

Lemma 3 (Commutativity and associativity of joint entropy)**.** *If $X : \Omega \to S$, $Y : \Omega \to T$, and $Z : \Omega \to U$ are random variables, then $\mathbb{H}[X, Y] = \mathbb{H}[Y, X]$ and $\mathbb{H}[X, (Y, Z)] = \mathbb{H}[(X, Y), Z]$.*

Proof. Set up an injection from (X, Y) to (Y, X) and use Theorem 2 for the first claim. Similarly for the second claim. □

Note: it may happen that E has zero measure. In which case, the ambient probability measure should be replaced with a zero measure. (In our formalization we achieve this by working with arbitrary measures, and normalizing them to be probability measures if possible, else using the zero measure. Conditioning is also formalized using existing Mathlib definitions.)

Definition 4 (Conditional entropy)**.** *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, the conditional entropy $\mathbb{H}[X|Y]$ is defined as*

$$\mathbb{H}[X|Y] := \sum_{y \in Y} \mathbb{P}[Y = y]\mathbb{H}[(X|Y = y)].$$

Lemma 5 (Chain rule)**.** *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, then*

$$\mathbb{H}[X, Y] = \mathbb{H}[Y] + \mathbb{H}[X|Y].$$

Proof. Direct computation. □

Lemma 6 (Alternative formulae for mutual information)**.** *With notation as above, we have*

$$\mathbb{I}[X : Y] = \mathbb{I}[Y : X]$$
$$\mathbb{I}[X : Y] = \mathbb{H}[X] - \mathbb{H}[X|Y]$$
$$\mathbb{I}[X : Y] = \mathbb{H}[Y] - \mathbb{H}[Y|X]$$

Proof. Immediate from Lemmas 3, 5. □

Lemma 7 (Nonnegativity of mutual information)**.** *We have $\mathbb{I}[X : Y] \geq 0$.*

Proof. An application of jensen's inequality and Theorems 6 and 21. □

Corollary 8 (Conditioning reduces entropy)**.** *With notation as above, we have $\mathbb{H}[X|Y] \leq \mathbb{H}[X]$.*

Proof. Combine Theorem 7 with Theorem 6. □

Corollary 9 (Submodularity)**.** *With three random variables X, Y, Z, one has $\mathbb{H}[X|Y, Z] \leq \mathbb{H}[X|Z]$.*

Proof. Apply the "averaging over conditioning" argument to Theorem 8. □

Corollary 10 (Alternate form of submodularity)**.** *With three random variables X, Y, Z, one has*

$$\mathbb{H}[X, Y, Z] + \mathbb{H}[Z] \leq \mathbb{H}[X, Z] + \mathbb{H}[Y, Z].$$

Proof. Apply Theorem 9 and Theorem 5. □

Lemma 11 (Vanishing of mutual information)**.** *If X, Y are random variables, then $\mathbb{I}[X : Y] = 0$ if and only if X, Y are independent.*

Proof. An application of the equality case of Jensen's inequality and Theorem 21. □

Lemma 12 (Alternate formula for conditional mutual information)**.** *We have*

$$\mathbb{I}[X : Y|Z] := \mathbb{H}[X|Z] + \mathbb{H}[Y|Z] - \mathbb{H}[X, Y|Z]$$

and

$$\mathbb{I}[X : Y|Z] := \mathbb{H}[X|Z] - \mathbb{H}[X|Y, Z].$$

Proof. Routine computation. □

Definition 13 (Conditionally independent random variables)**.** *Two random variables $X : \Omega \to S$ and $Y : \Omega \to T$ are conditionally independent relative to another random variable $Z : \Omega \to U$ if $P[X = s \wedge Y = t|Z = u] = P[X = s|Z = u]P[Y = t|Z = u]$ for all $s \in S, t \in T, u \in U$. (We won't need conditional independence for more variables than this.)*

Lemma 14 (Vanishing conditional mutual information)**.** *If X, Y, Z are random variables, then $\mathbb{I}[X : Y|Z] = 0$ iff X, Y are conditionally independent over Z.*

Proof. Immediate from Theorem 11 and Theorem 13. □

Corollary 15 (Entropy of conditionally independent variables)**.** *If X, Y are conditionally independent over Z, then*

$$\mathbb{H}[X, Y, Z] = \mathbb{H}[X, Z] + \mathbb{H}[Y, Z] - \mathbb{H}[Z].$$

Proof. Immediate from Theorem 14 and Theorem 12. □

One random variable is said to be a copy of another if they have the same distribution.

Lemma 16 (Copy preserves entropy)**.** *If X' is a copy of X then $\mathbb{H}[X'] = \mathbb{H}[X]$.*

Proof. Immediate from Theorem 1. □

Lemma 17 (Existence of independent copies)**.** *Let $X_i : \Omega_i \to S_i$ be random variables for $i = 1, \ldots, k$. Then if one gives $\prod_{i=1}^k S_i$ the product measure of the laws of X_i, the coordinate functions $(x_j)_{j=1}^k \mapsto x_i$ are jointly independent random variables which are copies of the $X_1, \ldots, X_k$.*

Proof. Explicit computation. □

Definition 18 (Ruzsa distance)**.** *Let X, Y be G-valued random variables (not necessarily on the same sample space). The* Ruzsa distance *$d[X; Y]$ between X and Y is defined to be*

$$d[X; Y] := \mathbb{H}[X' - Y'] - \mathbb{H}[X']/2 - \mathbb{H}[Y']/2$$

where X', Y' are (the canonical) independent copies of X, Y from Theorem 17.

Lemma 19 (Ruzsa distance in independent case)**.** *If X, Y are independent G-random variables then*

$$d[X;Y] := \mathbb{H}[X-Y] - \mathbb{H}[X]/2 - \mathbb{H}[Y]/2.$$

Proof. Immediate from Theorem 18 and Lemmas 2, 16. □

Lemma 20 (Existence of conditional independent trials)**.** *For X, Y random variables, there exist random variables X_1, X_2, Y' on a common probability space with $(X_1, Y'), (X_2, Y')$ both having the distribution of (X, Y), and X_1, X_2 conditionally independent over Y' in the sense of Theorem 13.*

Proof. Explicit construction. □

2 Applications of Jensen's inequality

In this chapter, h denotes the function $h(x) := x \log \frac{1}{x}$ for $x \in [0,1]$.

Lemma 21 (Concavity)**.** *h is strictly concave on $[0, \infty)$.*

Proof. Check that h' is strictly monotone decreasing. □

Lemma 22 (Balog-Szemerédi-Gowers)**.** *Let A, B be G-valued random variables on Ω, and set $Z := A + B$. Then*

$$\sum_z \mathbb{P}[Z=z] d[(A|Z=z); (B|Z=z)] \leq 3\mathbb{I}[A:B] + 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]. \quad (1)$$

Proof. Let (A_1, B_1) and (A_2, B_2) (and Z', which by abuse of notation we call Z) be conditionally independent trials of (A, B) relative to Z as produced by Theorem 20, thus (A_1, B_1) and (A_2, B_2) are coupled through the random variable $A_1 + B_1 = A_2 + B_2$, which by abuse of notation we shall also call Z.

Observe from Theorem 19 that the left-hand side of (1) is

$$\mathbb{H}[A_1 - B_2|Z] - \mathbb{H}[A_1|Z]/2 - \mathbb{H}[B_2|Z]/2. \quad (2)$$

since, crucially, $(A_1|Z=z)$ and $(B_2|Z=z)$ are independent for all z.

Applying submodularity (Theorem 10) gives

$$\begin{aligned} &\mathbb{H}[A_1 - B_2] + \mathbb{H}[A_1 - B_2, A_1, B_1] \\ &\quad \leq \mathbb{H}[A_1 - B_2, A_1] + \mathbb{H}[A_1 - B_2, B_1]. \end{aligned} \quad (3)$$

We estimate the second, third and fourth terms appearing here. First note that, by Theorem 15 and Theorem 2 (noting that the tuple $(A_1 - B_2, A_1, B_1)$ determines the tuple (A_1, A_2, B_1, B_2) since $A_1 + B_1 = A_2 + B_2$)

$$\mathbb{H}[A_1 - B_2, A_1, B_1] = \mathbb{H}[A_1, B_1, A_2, B_2, Z] = 2\mathbb{H}[A, B] - \mathbb{H}[Z]. \quad (4)$$

Next observe that

$$\mathbb{H}[A_1 - B_2, A_1] = \mathbb{H}[A_1, B_2] \leq \mathbb{H}[A] + \mathbb{H}[B]. \quad (5)$$

Finally, we have

$$\mathbb{H}[A_1 - B_2, B_1] = \mathbb{H}[A_2 - B_1, B_1] = \mathbb{H}[A_2, B_1] \leq \mathbb{H}[A] + \mathbb{H}[B]. \quad (6)$$

Substituting (4), (5) and (6) into (3) yields

$$\mathbb{H}[A_1 - B_2] \leq 2\mathbb{I}[A : B] + \mathbb{H}[Z]$$

and so by Theorem 8

$$\mathbb{H}[A_1 - B_2|Z] \leq 2\mathbb{I}[A : B] + \mathbb{H}[Z].$$

Since

$$\begin{aligned}\mathbb{H}[A_1|Z] &= \mathbb{H}[A_1, A_1 + B_1] - \mathbb{H}[Z] \\ &= \mathbb{H}[A, B] - \mathbb{H}[Z] \\ &= \mathbb{H}[Z] - \mathbb{I}[A : B] - 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]\end{aligned}$$

and similarly for $\mathbb{H}[B_2|Z]$, we see that (2) is bounded by $3\mathbb{I}[A : B] + 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]$ as claimed. □